HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained **Java Spring Boot** REST APIs for enterprise applications, handling approximately **1,500 requests per minute** during peak hours.
- Refactored and optimized **SQL queries** across PostgreSQL and Oracle databases, reducing data retrieval times for critical reporting functions by an average of **18%**.
- Automated data validation and report generation processes using **Python scripting**, successfully processing over **50,000 records daily** and reducing manual effort by **30%**.
- Designed and implemented **valuation models** for prospective energy contracts, incorporating quantitative methods to assess standalone and portfolio-level risk for new deal structures.
- Conducted in-depth **quantitative and statistical analysis** of power, gas, and carbon markets using Python and SQL to inform pricing strategies and deal structuring for new contracts.
- Collaborated with **Origination and Trading teams** to provide commercial insight and independent views on pricing and structuring for live energy deals, ensuring alignment with market conditions.
- Containerized multiple microservices using **Docker** and deployed them to an **AWS EKS cluster**, streamlining deployment pipelines and improving resource utilization.

SKILLS

Programming Languages – Java, Python, SQL, R, DAX

Frameworks & Libraries – Spring Boot, pandas, NumPy, scikit-learn, Flask, FastAPI

Databases – PostgreSQL, MySQL, Oracle

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch)

DevOps & Tools – Docker, Jenkins, Git, Jira, Confluence, Splunk, Dynatrace

Analytics & BI – Power BI, Tableau, Power Query, data modelling, data warehousing, statistical analysis, A/B testing

PROJECTS

Energy Contract Pricing & Risk Analysis Engine

- Developed a **Python-based analytical engine** to price complex energy contracts, including wind and solar balancing agreements and PPAs, integrating market data feeds.
- Implemented **scenario analysis and sensitivity testing** within the pricing models to quantify potential risks and opportunities associated with different deal structures.
- Designed and built interactive **dashboards in Power BI** to visualize pricing outcomes, risk exposures, and portfolio construction insights for senior management and Origination teams.

Automated Transaction Reconciliation System (HCL Technologies)

- Engineered an automated reconciliation system using **Java Spring Boot** and **PostgreSQL** to match high-volume transaction data from disparate sources, reducing manual reconciliation effort by **45%**.
- Developed **Python ETL scripts** to process and transform raw transaction logs, handling approximately **100,000 records per day** and preparing data for the reconciliation engine.
- Implemented robust error handling and logging mechanisms to identify and flag discrepancies, ensuring data integrity and supporting audit trail requirements for financial operations.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)